

4. Therapeutic drug monitoring

4.1.Principles and applications

Drug Therapy Monitoring Definition

Drug therapy monitoring, also known as Therapeutic Drug Monitoring (TDM), is a means of monitoring drug levels in the blood.

Purpose

TDM is employed to measure blood drug levels so that the most effective dosage can be determined, with toxicity prevented. TDM is also utilized to identify noncompliant patients (those patients who, for whatever reason, either cannot or will not comply with drug dosages as prescribed by the physician).

Precautions

Because so many different factors influence blood drug levels, the following points should be taken into consideration during TDM:

- The age and weight of the patient;
- The route of administration of the drug;
- The drug's absorption rate, excretion rate, delivery rate, and dosage;
- Other medications the patient is taking;
- Other diseases the patient has;
- The patient's compliance regarding the drug treatment regimen; and
- The laboratory methods used to test for the drug.

REASONS FOR MONITORING DRUG TREATMENT

- a) To see whether there is a therapeutic response
- b) To assess drug toxicity
- c) To assess compliance

Principle of Therapeutic Drug monitoring test

- The test should be safe, simple, precise, and validated
- The distribution of test values in the exposed population should be known and suitable cut-off values established
- The test should be acceptable to treated patients
- A strategy in the face of a positive monitoring test should be agreed

The response to positive tests

- An effective intervention should exist
- This early intervention should make the outcome better than it would have been with delayed intervention
- Evidence for the intervention should be robust

The monitoring strategy

- The strategy should reduce morbidity or mortality from the adverse effect
- The strategy should be acceptable to patients and professionals
- Benefits of monitoring should outweigh the physical and psychological harm
- The cost of monitoring should be proportionate
- A system for assuring the standards of the monitoring programme should exist
- Possibility of reducing or removing risks of adverse effects by selection of drug or dosage, or by pretreatment detection of susceptible people, should have been fully explored

Pharmacokinetic differences in drug handling between patients produce wide variability in serum drug concentrations, but dosage adjustments are generally only required if there is serious organ damage such as severe renal impairment and renal failure. Dosage requirements of individual patients can also vary significantly if a drug has a narrow therapeutic index. However, therapeutic effect is not always easily measured and toxic effects may be non-specific. In these circumstances, assuming the relationship between the drug concentration and the response is known; measuring the concentration of the drug in the patient's blood can prove useful information. But in our country, Ethiopia, Therapeutic drug Monitoring is done clinically, due to absence of the materials for measuring drug concentration in drug and different body parts

Therapeutic drug monitoring (TDM) refers to the individualisation of dosage by maintaining plasma or blood drug concentrations within a target range (therapeutic range, therapeutic window). This range represents plasma concentrations generally associated with optimal effect and a low incidence of toxicity. It is a guide to therapy but must be used in conjunction with clinical response when assessing and adjusting dosage regimens.

TDM is costly in addition to being unpleasant and inconvenient for the patient. Therefore the reason for monitoring and the impact of additional information on clinical decisions must be carefully considered, especially in resource scarce facilities like.

TDM may be indicated in the following situations:

- When initiating therapy or adjusting dose
- In the critically ill or patients with rapidly changing physiological status eg acute renal failure
- To confirm toxicity/lack of effect in a poorly controlled patient
- Where a drug interaction is suspected
- As a measure of patient compliance

Drug concentrations are usually measured in whole blood or serum, although saliva, which gives a measure of the unbound drug concentration, may be useful when blood samples are difficult to collect. For example, measuring phenytoin concentrations in children.

The correct timing of sampling is important for accurate TDM. For most drugs the “target” concentrations are based on steady state samples taken at specific times after the dose. In most clinical situations, the attainment of steady state can be assumed 3 to 5 half-lives after change in dose or initiation of therapy provided that the dose and clearance remain constant (unless loading doses are given). For most drugs, plasma samples should be taken just before the next dose (trough) as these levels are less likely to be influenced by absorption and distribution problems.

Assays measure drug that is both bound and unbound to plasma protein, but it is only the unbound drug that interacts with the receptor to produce a response. If binding is changed by disease states, displacement by another drug or non- linearity in protein binding, the interpretation of total plasma or blood drug concentrations must be modified to prevent inappropriate dose adjustments resulting in toxicity.

Another factor that should be considered when interpreting drug concentrations is

whether the drug has active metabolites, as metabolites which may not be measured can contribute to the therapeutic response.

4.1. Drugs requiring therapeutic drug monitoring

4.1.1. Anti-infectives

Many anti-infective agents have a wide therapeutic range, which means that standard doses can be used. However, with some antibiotics, toxicity is associated with persistently high concentrations or treatment failure with persistently low concentrations.

Aminoglycoside antibiotics

Aminoglycoside antibiotics eg gentamicin display *concentration-dependent* toxicity. The aim of treatment is to achieve high peak concentrations but allow the concentration to fall to low levels between doses. For this reason, most hospitals now use once daily dosing of 5 -7 mg/kg and various methods of monitoring levels are implemented.

Glycopeptide antibiotics

Glycopeptide antibiotics eg vancomycin and teicoplanin display *time-dependent* toxicity, so the aim is to maintain concentrations above the minimum inhibitory concentration (MIC) for most of the dosage interval. Different studies recommend trough concentrations between 10 and 20 mg/L for vancomycin, whereas some hospitals use constant rate infusions aiming to maintain concentrations between 15 and 25mg/L. Teicoplanin concentrations are not measured routinely but can be useful if under dosing is suspected. The data sheet recommends that in patients with impaired renal function, trough plasma teicoplanin concentrations should be monitored periodically after the first week of therapy and the dosage adjusted to prevent trough concentrations exceeding 30 µg/mL in patients with septic arthritis or 15 µg/mL in other cases, with a minimum of 10 µg/mL.

Anti-fungal drugs

Amphotericin concentrations are rarely measured. Flucytosine concentrations should

be at least 20-25 µg/ml and transiently not more than 100-120 µg/ml.

Prolonged concentrations of over 100 µg/ml must be avoided because of the increased risk of side effects. There is some evidence that trough concentrations of itraconazole above 0.5mg/L are desirable so monitoring of this drug could become more common.

Anti-retroviral drugs

Unexplained treatment failures and a high incidence of drug interactions have led to an interest in the monitoring of anti-retroviral drug concentrations. The protease inhibitors have received particular attention because of their complex and variable pharmacokinetics and the recognised association between trough plasma concentrations and therapeutic response.

4.1.1. Anti-convulsants

Valproate

Sodium valproate is highly bound to plasma protein, and at therapeutic plasma concentrations, saturates plasma protein binding sites causing the amount of free drug to rise rapidly, out of proportion to any increase in dosage. Correlation between the dose and plasma levels of valproate have been poor and levels are not a useful index of efficacy. Optimum dosage is mainly determined by seizure control and routine measurement of plasma levels is unnecessary. However, a method for measurement of plasma levels is available and may be helpful where there is poor control or side effects are suspected and to check adherence.

Carbamazepine

Carbamazepine has an active metabolite which makes interpretation of the plasma concentrations difficult. Although correlations between dosage and plasma levels of carbamazepine, and between plasma levels and clinical efficacy or tolerability are rather tenuous, carbamazepine concentrations are occasionally monitored. It is important to remember that carbamazepine induces its own metabolism, so levels taken before 4 weeks may not reflect steady state. The steady-state plasma concentrations of carbamazepine considered as 'therapeutic range' vary considerably interindividually: for the majority of patients a range between 17 to 50 micromol/L

has been reported. Concentrations of carbamazepine-10,11-epoxide (pharmacologically active metabolite) are about 30% of carbamazepine levels.⁶

Phenytoin

Phenytoin monitoring remains useful because its non-linear kinetics make dose adjustment difficult.¹ Phenytoin plasma concentrations of 10 to 20mg/L are generally accepted as therapeutic. Plasma concentrations between 5 to 10mg/L can be therapeutic for some patients, but concentrations less than 5mg/L are not likely to be effective. In most patients maintained at steady dosage, stable phenytoin serum levels are achieved. There may be wide interpatient variability in phenytoin serum levels with equivalent dosages. Patients with unusually low levels may be noncompliant or hypermetabolisers of phenytoin. Care is required in patients with low albumin or renal failure because the consequent reduction in plasma protein binding can lead to misinterpretation of total concentration measurements. Phenytoin plasma concentrations of 10 to 20mg/L correspond to an unbound drug concentration of 1-2 mg/L (fraction unbound, f_u is normally 0.1). If f_u is increased to 0.2, as, for example, in renal disease, the target unbound concentration is still 1-2 mg/L, but the therapeutic range for total drug is 5-10 mg/L.

A number of phenytoin side effects, such as gingival hyperplasia, folate deficiency and peripheral neuropathy do not appear to be directly related to plasma phenytoin concentrations. Conversely, CNS side effects do correlate with plasma concentration.

The newer anticonvulsants have wider therapeutic ranges, so monitoring is rarely performed in practice. Nevertheless, a tentative target range of 3 to 14mg/L has been proposed for lamotrigine, and there is some evidence that adverse effects of lamotrigine may be concentration dependent.

Different books recommend the following guidelines for monitoring anti- convulsant therapy:

- monitoring initial stabilisation or change of dose – phenytoin, carbamazepine, phenobarbitone

- suspected toxicity – all drugs
- suspected non-compliance – all drugs
- failure to control seizures – all drugs
- ongoing routine monitoring – phenytoin only, and even this may not be essential.

4.1.2. Cardiovasculars

Digoxin

Digoxin has a large volume of distribution, and absorption and distribution of drug takes a long time. For this reason, samples taken less than 6 hours after an oral dose cannot be accurately interpreted as the plasma concentration will be erroneously high.

Another common error with digoxin is to ignore the effect of decreased renal function in patients on long-term therapy. As the major route of digoxin elimination is renal excretion of the unchanged medicine, renal impairment can lead to accumulation of drug and have toxic consequences.

Although there is considerable variation between patients, plasma digoxin concentrations of 1 to 2nmol/L are generally considered to be within the therapeutic range. Digoxin toxicity is more commonly associated with serum digoxin concentration greater than 2nmol/L; however, toxicity may occur with lower digoxin serum concentrations. In deciding whether a patient's symptoms are due to digoxin, the clinical state together with the serum potassium level and thyroid function are important factors.

4.1.3. Immunosuppressants

Many of the immunosuppressive drugs have a narrow therapeutic index in addition to exhibiting pharmacokinetic variability; both inter-patient and intra-patient. Sub-therapeutic levels and wide fluctuations in drug levels are risk factors for graft rejection and decreased graft survival. TDM is essential in optimizing the patient's immunosuppressive drug regimen to minimize the risk of rejection and dose-related adverse effects.

The therapeutic range for immunosuppression can vary according to transplant centre protocols and may be specific to the organ transplanted, time since transplant and indication for transplant.

Cyclosporin

The most common adverse effect associated with cyclosporin is renal dysfunction, which usually improves when the dose is decreased or drug is discontinued although can lead to irreversible renal failure. For this reason cyclosporin doses and target concentrations are often lowered in patients who show early signs of nephrotoxicity. Other dose-related adverse effects include neurotoxicity (headache, tremors, paresthesias, seizures) and hypertension.

Cyclosporin has limited and highly variable bioavailability, and although the most widely used method for monitoring cyclosporin therapy is to sample the trough concentration, this approach does not always correlate well with the patient's exposure to drug. AUC is a much better predictor of response and the cyclosporin concentration at 2 hours after administration (C_2) is the most accurate single sample marker for AUC. There is a 15 minute period before and after the 2 hour time point, during which the C_2 sample can be taken to remain within an acceptable margin of error.

Tacrolimus

Tacrolimus also has a low bioavailability, and is associated with dose-related nephrotoxicity, neurotoxicity, post-transplant diabetes mellitus and hypertension. Tacrolimus has a more consistent relationship between trough concentration and AUC, so therapy is usually based on trough measurements.

4.1.4. Other Drugs

Anti-cancer drugs

Although chemotherapeutic agents have narrow therapeutic ranges, concentrations are not routinely monitored because of a lack of data on concentration-effect relationships. One exception is methotrexate where folinic acid rescue is based on

monitoring the methotrexate concentration 24-48 hours after the high dose therapy and continuing until concentrations are below 1.5 $\mu\text{mol/L}$.

Psychiatric drugs

Target ranges have been identified for a number of antidepressants and antipsychotics, many of which are subject to genetic polymorphism resulting in significant differences in dosage requirements eg nortriptyline. The therapeutic ranges for imipramine and amitriptyline are based on the combined concentrations of parent drug and active metabolite (desipramine and nortriptyline respectively).

Lithium

Lithium has a narrow therapeutic range and exhibits fairly predictable pharmacokinetic behaviour; it is an excellent candidate for routine therapeutic monitoring. The incidence and severity of acute adverse effects increase significantly with higher plasma concentrations, and there are also a variety of long-term adverse effects that are commonly associated with high plasma concentrations eg fine tremor, weight gain, thyroid abnormalities and cognitive dulling. The target range for acute treatment of mania is 0.8 to 1.2 mmol/L, whereas a range of 0.6 to 0.8 mmol/L is used for maintenance therapy. The ideal time for sampling is just before the dose or at least 12 hours after the previous dose, and steady state levels should be attained within 3 to 5 days. Changes in renal function, fluid and electrolyte balance and interacting drugs can alter drug concentrations as can irregular dosing, poor compliance and missed doses. These factors need to be considered as an explanation for poor therapeutic response or toxicity.

Theophylline

Theophylline is principally cleared by hepatic metabolism, therefore dosage requirements vary widely and drug interactions remain a major concern. The half-life of theophylline is increased and clearance decreased in the elderly and in patients with congestive heart failure, chronic obstructive pulmonary disease, or liver disease. The clearance in premature infants and neonates is also reduced, whereas the serum half-life is decreased and clearance increased

in cigarette or marijuana smokers. Bronchodilator effects have been demonstrated within the normal target range of 10 to 20 mg/L but lower concentrations are associated with anti-inflammatory and steroid sparing effects. Consequently, there is some support for reducing the target range to 5 to 15mg/L which may reduce the incidence of toxic effects. Theophylline in neonates (but not in adults) is converted to caffeine, so the therapeutic range for theophylline in neonatal apnoea is lower: 6 to 12 mg/L (allowing for the contribution of caffeine). The average theophylline half-life is approximately 8 hours; therefore routine monitoring of plasma concentrations can usually begin approximately 24 hours after the initiation of therapy or change in dose. In patients receiving nonsustained-release or liquid dosage forms, monitoring is most reliable when trough levels are obtained. In patients receiving sustained-release theophylline, the time of sampling is less critical. Trough concentrations are recommended, but samples taken at the midpoint of the dosing interval may also be acceptable.

Commonly monitored drugs and its recommended sampling time

Medicine (Therapeutic range)	Elimination $t_{1/2}$	Approximate time to C_{ss}	Recommended sampling time
Carbamazepine (17-50 μ mol/L)	12-17 hours (adult; chronic therapy)	2-4 days (adult)	Trough
Cyclosporin (150-350 ng/ml)	6-24 hours	5 days	Trough

Digoxin (1-2nmol/L)	38 hours (longer if renally impaired)	7 days (longer if renally impaired unless loading dose given)	Trough or >6 hours after oral dose
Lithium (0.8-1.2 mmol/L acute)(0.6-0.8 mmo/L	18-24 hours	5 days	Trough or >12 hours after dose
Phenytoin (10-20mg/L)	Dose dependent; usually 20-30 hours	5 days	Trough
Tacrolimus (5-20mg/L)	8-12 hours	24-36 hours	Trough
Theophylline (5-20mg/L)	6-9 hours (highly variable)	2 days	Trough